SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 1 – SHORT ANSWER TEST (5 QUESTIONS)

Course Code:	ITA0509	**Course Title:**	**Computer Vision**
CO Assessed:	CO1	**Assessment:**	Assessment 1: Short Answer Test
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO1 Statement:	***Apply the fundamental concepts, image formation models, and image processing techniques to analyze and solve real-world Computer Vision problems.***		
Student Name:	Vishnu Priya. G	**Reg. No.:**	192425309
Section / Batch:	B-Tech AIML	**Date:**	05/10/26

Code of Conduct (192425309)

I Vishnu Priya G *(NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student:

QUESTIONS: 5

Total Marks:50

Annexure A

Short Answer Test

Q1. Smart Traffic Monitoring System

A smart city is planning to install cameras at major traffic junctions to monitor vehicles, pedestrians, and traffic conditions. The captured images may vary in resolution, lighting, and camera position. **Apply the fundamental concepts of Computer Vision to design a suitable processing pipeline for this traffic monitoring system. Explain the major stages from image acquisition to object detection and classification.**

Q2. Automated Attendance System

A college wants to implement an automated attendance system using cameras that capture students sitting inside classrooms. The images may contain different lighting conditions, face orientations, shadows, and multiple students.

Analyze how image representation, preprocessing, feature extraction, and face recognition can be applied to develop the proposed attendance system. Discuss the major challenges that may affect its accuracy.

Q3. Medical Image Analysis System

A hospital plans to use a Computer Vision system to assist doctors in analyzing X-ray images for identifying abnormal regions. The images may contain noise, poor contrast, and different brightness levels.

Evaluate the importance of image resolution, intensity, preprocessing, segmentation, and feature extraction in this application. Explain how these factors can influence the accuracy and reliability of the system.

Q4. Security Surveillance System

A shopping mall wants to develop an intelligent surveillance system using cameras installed at entrances, corridors, and parking areas. The system must identify people and objects under different lighting and crowded conditions.

Analyze how the fundamental stages of a Computer Vision system can be used for this application.

Explain the effects of illumination, occlusion, camera position, and image quality on the final results.

Q5. Smart Agriculture Monitoring

A farming organization wants to use drone cameras to identify healthy plants, damaged plants, weeds, and disease-affected areas in large agricultural fields. The captured images may contain shadows, different lighting conditions, and variations in plant size.

Create a suitable Computer Vision pipeline for this application using image acquisition, preprocessing, segmentation, feature extraction, and classification. Justify the purpose of each stage.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, in-depth4, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature:	Signature:	Signature:
Date: 3/10/26	Date:	Date:

Smart Traffic Monitoring System

A Suitable Computer Vision pipeline for a Smart traffic monitoring Systems Consists of the following Stages

1. Image Acquisition

Cameras installed at traffice junctions continuosly Capture images or video of Vehicles, Pedestrians and Road Conditions. Higher resolution Cameras Provide more detailed information.

2. Preprocessing

Since Images have poor lighting, noise or different resolutions, Preprocessing is Performed using resizing, noise reduction, brightness adjustment, Contrast enhancement and normalization

3. Segmentation

Import regions such as roads, vehicles, Pedestrians and traffic Signals are Separated from the background

4. Object Detection

Detection algorithms Such as YOLO or fast-er R-CNN locate Vehicles, pedestrians, bicycles.

5. Classification

Each detected object is classified into categories Such as Car, bus, truck or motorcycle

6. Output

The System can Calculate traffic density, Vehicle Count, Pedestrian movement and Congestion levels

2. Automated Attendance System

An automated attendance System can use Computer Vision and face recognition to identify students from Classroom images.

1. Representation of Images

The Captured classroom image is represented as a matrix of pixels. Each pixel Contains intensity or RGB Color values.

2. Preprocessing

Images are resized and normalized. Noise reduction, brightness Correction, Contrast enhancement, and face alignment are applied to improve image quality.

3. Face Detection

The System detects individual faces from the classroom image using methods Such as Haar Cascade, MTCNN or deep learning-based face detectors

4. Feature Extraction

Important facial characteristics Such as eyes, nose, mouth, face shape and deep facial embeddings are extracted.

5. Face Recognition

Extracted features are Compared with previously Stored Student face features in the database

6. Attendance Marking:

When a face matches a registered Student, the Student's attendance is automatically recorded.

Medical Image Analysis System

Computer Vision can assist doctors by identifying abnormal regions in X-ray image

1. Image Resolution

High resolution X-rays contain more detailed information about bones, tissues and abnormalities. Very low resolution may hide small abnormalities

2. Intensity

Pixel intensity represents the brightness of different structures. Differences in intensity help distinguish normal and abnormal regions

3. Preprocessing

Noise removal, contrast enhancement, normalization and brightness correction improve image quality. Techniques such as Gaussian filtering and histogram equalization may be used.

4. Segmentation

The X-ray is divided into important regions so that organs, bones, tissues, or suspicious areas can be analyzed separately.

5. Feature extraction

Features such as shape, texture, edges, intensity patterns and abnormal structures are extracted.

6. Classification

A machine learning or deep learning model classifies the region as normal or abnormal.

4. Security Surveillance System

An Intelligent Security Surveillance System can follow these Computer Vision Stages

1. Image Acquistion

Cameras installed at entrances, corridors and parking areas capture Continuous video

2. Preprocessing

Noise reduction, resizing, Contrast enhancement, brightness correction and normalization improve the captured frames.

3. Segmentation

Foreground Objects Such as people, vehicles and Suspicious objects are Separated from the background

4. Feature Extraction

Features Such as Shape, texture, facial characteristics, body appearance, and movement patterns are extracted.

5. Object Detection

Computer Vision models detect people, bags, vehicles, and other objects.

6. Classification and Tracking

Detected objects are classified and tracked across multiple video frames

Conclusion:-

Proper Camera placement and image Preprocessing are essential for reliable Surveillance.

Smart Agriculture Monitoring

A Suitable Computer Vision pipeline for Smart Agriculture is

Image Acquisition → Preprocessing → Segmentation → Feature Extraction → Classification → Result

Image Acquisition

Drone Cameras Capture Images of large agricultural fields. High resolution images help observe individual plants and damaged regions.

2. Preprocessing

Images are resized, denoised, normalized. This Produces Consistent image quality

3. Segmentation

Plant regions are Separated from Soil, weeds, background. Diseased leaves or damaged regions can also be isolated.

4. Feature extraction

Important features Such as leaf color, Shape, texture, edges, Spots are extracted.

5. Result

The System produces a field map showing healthy and affected agriculture regions